

Counting on Working Memory When Learning to Count and to Add: A Preschool Study

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In this study, the author aimed at measuring how much limited working memory capacity constrains early numerical development before any formal mathematics instruction. To that end, 4- and 5-year-old children were tested for their memory skills in the phonological loop (PL), visuo-spatial sketchpad (VSSP), and central executive (CE); they also completed a series of tasks tapping their addition and counting skills. A general vocabulary test was given to examine the difference between the children's numerical and general vocabulary. The results indicated that measures of the PL and the CE, but not those of the VSSP, were correlated with children's performance in counting, addition and general vocabulary. However, the predictive power of the CE capacity was significantly stronger than that of the PL capacity. Poor CE capacity should thus be taken into consideration when identifying children at risk of experiencing learning disabilities.

Keywords: vocabulary, counting, addition, working memory, mathematics

This study examines the numerical learning that takes place in preschoolers before any formal mathematical instruction and, more precisely, the learning of the names of the numbers, the development of the ability to recite number names in the correct order, the gradual elaboration of the counting string, and the use of counting abilities to perform simple calculations. I examined whether variations in the capacity of the children's working memory could explain individual differences in acquiring these skills.

Working Memory

Working memory is known to play an important role in many cognitive processes and to be a key factor in their development (e.g., Sodian, Schneider, & Schumann-Hengsteler, 2005). In Baddeley and Hitch's model (Baddeley, 1998; Baddeley & Hitch, 1974), the working memory (WM) is conceived of as a multicomponent system. It involves two slave systems, the *phonological loop* (PL) and the *visuo-spatial sketchpad* (VSSP), that are posited to store verbal and visuo-spatial information, respectively, and a *central executive* (CE) that is responsible for controlling the two slave systems and allocating resources between them. A fourth component, called the *episodic buffer*, has been added to the model to deal with the integration between this system and long-term

memory (Baddeley, 2000), but as it has not been studied in children, I did not consider it here.

Other theoretical frameworks also exist. For instance, Cowan (2001) takes the working memory system to be the activated portion of long-term memory and the focus of attention, together with a set of sensory buffers that hold information for very brief time intervals. In this sense, a distinction is drawn between passive storage systems that maintain the information without any sort of manipulation and more active processes that require storage in the context of competing processing requirements. The former correspond to the slave systems in Baddeley's model, whereas the latter allocate the main role to the CE. For the sake of simplicity, in this article I speak of PL measures when only passive storage of verbal material is required, VSSP measures when passive storage of visuo-spatial material is implied, and CE measures when complex tasks involving both the storage and the manipulation of information are required (e.g., repeating a sequence in the reverse order than that of presentation), without taking any position on the nature of the memory system (i.e., whether it is a separate system from, or an activated portion of, the long-term memory system).

Working Memory in Calculation

In the numerical domain, adult studies have shown a major involvement of the CE in both single-digit (De Rammelaere, Stuyven, & Vandierendonck, 2001; Hecht, 2002) and multidigit calculations (e.g., Fürst & Hitch, 2000), and the involvement of the PL in complex but not in simple arithmetic (Noël, Désert, Aubrun, & Seron, 2001). There is much less clear evidence for the use of the VSSP (for a review, see DeStefano & LeFevre, 2004).

Converging evidence also indicates the role of WM in calculation in primary school children. However, unlike adults who solve single-digit operations mainly by retrieving the solution from long-term memory, young children (up to Grade 3 or 4) use counting strategies or decomposition to solve such problems (e.g., Lemaire, Barret,

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Fayol, & Abdi, 1994). For instance, Fuson (1982) identified three basic counting algorithms to perform additions. In *counting all*, the child counts both terms starting from one. In *counting on from the first number*, the counting starts at the first number and continues by a number of units equal to the second number. Finally, in *counting min*, the child starts at the larger number and counts on by a number of units equal to the smaller number. Counting strategies also develop in terms of the type of support used by the child. Most 3-year-old children (and some 4-year-olds) use objects to represent and solve the problems (Fuson, 1982). Most 4- and 5-year-old children use a combination of finger counting and verbal counting (Siegler & Shrager, 1984). Fingers, like objects, are used to represent the problem, but also to help the child keep track of his or her counting (Fuson, 1982). The transition from finger to verbal counting depends primarily on the child's mental ability to keep track of the numbers that have already been counted and those that still need to be counted (Fuson, 1982).

The use of such counting strategies suggests a particularly strong involvement of the CE, which is supported by the available data. For instance, in a population of normal 7-year-old children, arithmetical performance has been shown to correlate with CE capacity (Bull & Johnston, 1997; Bull & Scerif, 2001; Bull, Johnston, & Roy, 1999; Gathercole & Pickering, 2000; Gathercole, Pickering, Knight, & Stegmann, 2004). Furthermore, poor memory, particularly in the CE component of memory, has been reported in poor solvers of arithmetical word problems (Passolunghi & Pazzaglia, 2005; Passolunghi & Siegel, 2004). Similarly, mathematically disabled children show low WM capacities for both the CE (Geary, Hoard, & Hamson, 1999; McLean & Hitch, 1999; Noël, 2004; Siegel & Ryan, 1989; Swanson, 1994) and, although less consistently, the PL components (Geary, Brown, & Samaranayake, 1991; Hitch & McAuley, 1991). A recent meta-analysis of these results can be found in Swanson and Jerman (2006).

Complex WM skills are also closely linked with children's academic progress (including their progress in mathematics) in the early school years (Gathercole & Pickering, 2000). In a short longitudinal study, Noël, Seron, and Trovarelli (2004) examined the link between WM capacities (i.e., the PL, measured by five verbal spans, and the CE, measured by the listening span) and calculation strategies in first-grade children. They showed that capacities measured at the beginning of Grade 1, especially those of the PL, were significant predictors of addition skills 4 months later. In particular, children with high PL capacities were more accurate in addition and used more mature strategies (such as retrieval from long-term memory) than children with poor PL skills. Conversely, children with limited WM capacity tended to use the counting-all algorithm more frequently, and still used their fingers to solve single-digit additions. From these data, Noël et al. (2004) hypothesized that limited WM resources prevent children from using more mature strategies for solving addition problems.

However, this important result was not replicated by Passolunghi, Vercelloni, and Schadee (2007). These authors tested a larger population of first-grade children ($N = 170$) at the beginning and end of the school year and showed that measures of the CE (backward digit and word spans and listening completion span) had a significant and specific impact on their performance in mathematics. However, this was not true of the PL (forward digit and word spans). McKenzie, Bull, and Gray (2003) used an interference paradigm in normal 6- and 8-year-old children. They found

that the accuracy of their addition decreased when the two slave systems were disrupted, although interference due to disruption of the VSSP was stronger than that due to disruption of the PL.

These studies of primary-school students indicate that, although the VSSP and PL appear to be of some importance in some studies, the role of the CE in learning mathematics is stronger. Few studies have considered the role of WM in the first stage of numerical development in preschool. To the best of my knowledge, only two studies have been published on this topic. In the first, Klein and Bisanz (2000) presented nonverbal calculations to 4-year-old children ($n = 48$): An array of tokens was placed on a mat and then covered; some tokens were then added or subtracted from the hidden array, and the children had to put the same number of tokens on their mat as the experimenter now had on his or her mat. The children's accuracy appeared to be strongly related to the "representational-set size" of the problem—that is, the maximum number of units that had to be held in working memory to solve a given problem (so that a representational-set size of five characterizes both the addition $3 + 2$ and the subtraction $5 - 2$). From this, the authors concluded that WM capacities are of major importance in the early development of arithmetic. However, the WM capacities of the children were not directly assessed in this study.

A second study was therefore conducted by Rasmussen and Bisanz (2005), which included measures of the CE (backward digit span and counting span), the VSSP (Corsi span), and the PL (digit span). These authors tested a total of sixty-three 5- and 6-year old children with two types of addition problems: nonverbal arithmetic (as described above) and verbal arithmetic problems presented aloud and requiring a verbal answer (e.g., "If you have five apples and I give you three more apples, how many apples have you got altogether?"). The children were better at solving nonverbal than verbal problems, and their performance on the nonverbal problems was related to measures of VSSP, whereas their performance on the verbal problems was correlated with one measure of CE (partial r^2 after controlling for age was .10 for the counting span, but was nonsignificant for the backward digit span). The presentation (and production) format of the problem thus seems to be very important at this age. Rasmussen and Bisanz argue that children solve nonverbal problems using a mental model based on object files.

The present study also deals with preschool children who have not been exposed to formal mathematical instruction. It aims to examine whether numerical development is related to WM capacities and to specify the WM components that are most involved in this relationship. As in Rasmussen and Bisanz's (2005) study, the capacities of the PL, VSSP, and CE are assessed. However, the numerical testing includes not only additions but also counting skills. Indeed, counting skills are the bases for the development of calculation strategies and are a good predictor of mathematical school achievement measured in Grade 1 ($r = .52, p < .01$) and even in Grade 4 ($r = .41, p < .01$; Krajewski & Schneider, 2008). Understanding the constraints of counting development is thus a very important issue.

Development of Counting

Very early on, children start reciting the names of the numbers. A typical structure emerges from the counting sequences that 3- and 4-year olds produce in repeated counting trials. The first

portion is stable over trials and corresponds to the conventional number sequence. It is followed by another stable portion that does not correspond to the conventional number sequence. Finally, an unstable (i.e., varying from trial to trial) and incorrect portion is produced (Fuson, Richards, & Briars, 1982). With development, the stable and conventional portion increases. Most children below the age of 3½ are still learning the sequence from 1 to 10, whereas many children older than 4½ are working on the numbers between 20 and 100.

Another process develops in parallel to this increase in the stable and conventional portion of the number sequence: the elaboration of the counting sequence. According to Fuson et al. (1982) and Fuson (1988), five levels of elaboration can be distinguished. At the *string level*, the words are a forward-directed, connected, and undifferentiated whole. At the *unbreakable list* level, the words are separated, but the sequence only exists as a forward-directed recitation and counting can only start from one. In the *breakable chain*, parts of the chain can be produced, starting from a number other than one. At that level, the child can count up from *a* to *b* (e.g., the child can count up from five to eight: “five, six, seven, eight”). At the *numerable chain* stage, the words become units in the numerical sense, and so sets of sequence words can be counted or matched. At that level, children are able to count up *n* steps from *a* (e.g., the child can count three steps from five: “six, seven, eight”). Finally, in the *bidirectional chain* stage, words can be produced easily and flexibly in either direction; the child can count up or down from any number and can easily shift direction.

Aim of This Study

This study aimed to explore the relationships between WM capacities and the development of both calculation and counting abilities. More specifically, I was interested in the development of the length of the counting sequence and its level of elaboration, which may be very important in performing additions.

No clear prediction can be made regarding the WM components that are more important in those numerical developments. On the basis of previous research, it could be hypothesized that learning the number words might be related to the child’s PL capacities, as they are a good predictor of the acquisition of more general vocabulary, especially between 4 and 5 years of age (Gathercole & Baddeley, 1993; Masoura, Gathercole, & Bablekou, 2004). However, a more recent longitudinal study has shown that poor PL capacity at 5 years of age was not associated with poor vocabulary or number skills 3 years later (Gathercole, Tiffany, Briscoe, Thorn, & the ALSPAC team, 2005). Rather, the results of this study suggest that the CE capacity might be more important (see Gathercole et al., 2004, 2005).

Retrieval of these number words in the correct order also requires the intervention of WM. No data on this topic are available for children, but several studies with adults suggest that keeping one’s place in a counting sequence involves the PL. Indeed, articulatory suppression increases repetition errors in a written backward-counting task (Healy & Nairne, 1985) and decreases accuracy when adults are asked to count elements of an array or numbers of events in a sequence (Logie & Baddeley, 1987). Concerning the level of elaboration of the counting sequence, I predicted that children with good CE capacity would have a better

development, as most of the tasks require monitoring, a function dedicated to the CE.

The predictions for additions are even more difficult. Research on primary school children strongly supports a link between CE capacity and performance on addition problems. However, in these studies the problems have usually been presented verbally or in Arabic numerals (i.e., using symbolic material). Rasmussen and Bisanz (2005) have shown the importance of the presentation format and have pointed out the specific role of the VSSP when nonverbal problems are presented through objects. In this study, I decided to present the problems to the children in such a way that they could be solved by a variety of methods. The problems were therefore presented both verbally and with pictures of objects, and tokens were available to the children to elaborate their answers. In these conditions, it is possible to explore how children deal with the situation: Do they use mental representations of the objects to solve the problems and rely mainly on their VSSP capacities, or do they develop strategies relying mainly on their CE (as older children do)?

Besides the accuracy rate in addition, I also observed the type of strategy and support the child used to solve the problems. With first grade children, Noël et al. (2004) have observed that good PL capacities are associated with more mature solving strategies and weaker need of external support for counting. However, they did not test the capacity of the VSSP, and only one measure of the CE was used. Fuson (1982) underlined the fact that external supports like fingers and objects are used to keep track of the counting process, a process typically devoted to the CE. Conversely, if children use visual representation of the problems, weak VSSP capacities could be associated with a strong need to use external support such as objects or fingers to keep track of the counting. Accordingly, one might hypothesize that all three components of the WM might be linked to some extent with the degree of maturity of the solving strategies of additions and the need to use external supports.

Finally, one could also consider a global numerical factor that would include the performance in all these very basic numerical tasks and examine how this would be related to the different WM components. To test these hypotheses, WM capacity, counting, and addition skills were assessed in 4- and 5-year-old children. Correlations and regression analyses were performed to identify the WM components that accounted for most variance of these numerical developments.

Method

Participants

A total of 80 French-speaking Belgian kindergarten children were tested. Thirty-eight of them were in the second year of kindergarten (K2). Their age ranged from 3 years 11 months to 5 years ($M = 4$ years 5 months ± 3.9 months). The others were in the third year of kindergarten (K3), and their ages varied from 4 years 10 months to 6 years 1 month ($M = 5$ years 5 months ± 3.9 months). There were 20 girls in K2 and 16 in K3.

Tasks

Children’s WM capacities, addition and counting skills as well as their nonverbal intelligence and general vocabulary were assessed.

Working memory. Under the rubric of working memory, the storage-based systems for PL and VSSP material were evaluated, as were the capacities of the CE.

PL. The capacity to store verbal material was assessed through three tasks that did not involve number words: (a) a one-syllable word span (Maillart, 2003), (b) a one-syllable food and animal word span, and (c) a repetition of nonwords. Nonword repetition is less contaminated than word span by the child's mental lexicon and has been shown to be a reliable predictor of vocabulary size (Gathercole, 2006). In the first two tasks, the examiner presented a series of words¹ to the child at the rate of one per second and asked him or her to repeat them in the order of presentation. The first level of difficulty included two words. This was followed by strings of three, four, and five words. There were three trials at each level of difficulty. If the child failed at two out of the three trials, the task was stopped. The corrected span was used as the dependent measure: This is the longest sequence for which two series were repeated correctly, plus .5 if one longer series was also correctly processed. In the nonword repetition task, items were made of consonant–vowel syllables. The task started with a two-syllable nonword to repeat (three trials) and was followed by longer nonwords of three, four, and five syllables (Poncellet, 2002). The same procedure and dependent variable as in the first two tasks was used.

In addition to these verbal spans, two other measures that have been suggested to explain developmental differences in immediate memory abilities were explored: the rate of decay of verbal information and the speed of retrieval of verbal information from long-term memory (see Gathercole, 1999). To assess the former (decay), children were asked to repeat nonwords of increasing length (i.e., those used in the nonword repetition task described above) after empty delays of 5 and 10 s. Again, three trials were used for the repetition of two-, three-, four-, and five-syllable nonwords, to be repeated after 5 or 10 s. To assess the speed of retrieval of verbal information from long-term memory, a speeded naming task was used. In this task, children had to name, as fast as they could, a series of 10 pictures (taken from Snodgrass & Vanderwart, 1980) printed on a sheet of paper (i.e., boots, car, cat, chair, dog, elephant, fish, flower, feet, hand). Each of them corresponded to a name that is usually acquired at 18 months of age (Chalard, Bonin, Meot, Boyer, & Fayol, 2003). The same task was repeated four times. On the basis of the median naming time, I computed the number of items named per second.

VSSP. The Corsi block was used to assess the VSSP. In this task, the child sits in front of nine identical cubes glued to a board. The examiner touches sets of cubes of increasing number, and the child has to imitate the tester after each set. The same procedure as for the verbal span measures was used here: three trials per length and calculation of the corrected span.

CE. Two complex span tasks were used to assess the children's CE capacities.² In the reverse-word span task, Maillart's (2003) one-syllable word series were presented to the child, who was asked to repeat the words in the reverse order of presentation. In the category-span task, one-syllable food or animal words were presented to the child, who was asked to repeat them by category, giving the foods first and then the animals. To help the child understand the instructions, trials with items drawn on cards were used first, and pictures of a forest and a plate were presented to the child to guide the recall of the animal and food names, respec-

tively. Here again, the procedure was the same as that used in the verbal span tasks—that is, three trials per length and calculation of the corrected span.

Counting skills. Several counting tasks were used with the children. They aimed to assess the highest number to which the child could count, the accurate stable portion of the counting sequence, and its level of elaboration.

Counting forward. In the count-forward task, the child was asked to count as high as he or she could. This task was repeated three times during the testing session (although not in succession). These repetitions allowed the child's best counting to be identified, as well as the accurate portion of the counting sequence that was stable across the three trials (*stable and conventional* [SC] counting).

Numerical fluency. The counting-forward task gives a first flavor of the extent of the child's numerical lexicon. However, in that task the number words have to be produced in the conventional order. It is possible that the child knows many other number words but does not produce them in the counting task because he or she does not know their order in the counting sequence. To assess this, a *number fluency* (N-fluency) task was proposed. Immediately after the forward-counting task, the child was asked whether he or she knew any other number names. The various number words produced in this task, as well as those produced in the three forward-counting trials (without taking account of the production order of the sequence), were recorded and constituted a measure of the child's numerical lexicon (repetition of the same number word counted only once).

Numerical decision. A *number decision* (N-decision) task was also included: Words were presented orally to the child, who had to decide whether or not they were number words. To explain the task to the child, the experimenter said:

I will say words and I'll ask you to tell me if they are numbers or not. Numbers are words that are used to count, such as one, two, or three. So, if I say "three," you say "Yes, it is a number," but if I say "scissors," you must say, "No, it is not a number because it is not used when counting."

Nineteen words were presented: 9 of them were number words (3 unit names, 3 tens names, and 3 teens names); the others were 5 words from ordered linear series (e.g., "Sunday, March") and 5 nonwords that were morphologically similar to number words (e.g., "huitante," made from *huit* [eight] and *-ante*, the typical ending of French multiples of 10 such as *quarante* [forty]); or "neufze," starting with *neuf* [nine] and ending with *-ze*, the typical ending of teens, such as *quinze* [fifteen]).

¹ Seven one-syllable words were used in the word span, corresponding to the English words *nest*, *bridge*, *bench*, *fire*, *hole*, *calf*, and *hand*. Words for the food–animal spans were *wine*, *rice*, *juice*, *pudding*, *plum*, *nut*, *apple*, *salt*, *honey*, *cabbage*, *bread*, *soup* and *milk* for what people eat and *rat*, *cow*, *chicken*, *lion*, *wolf*, *hind*, *calf*, *tiger*, *seal*, *fawn* and *cat* for the animals.

² I also included a measure of inhibition capacity: the day–night inspired by Gerstadt, Hong, & Diamond, 1994. Yet, as also reported by Rasmussen and Bisanz (2005), this measure proved to be nonsignificantly correlating with the other measures neither of WM nor with any of the measures of numerical or vocabulary development. Therefore, I do not report the data regarding this measure, as this did not bring any meaningful information.

Elaboration. The level of elaboration of the counting sequence was assessed through a series of tasks in which the child was asked to (a) count up to 6 and then up to 9; (b) count up from 3 and then from 7 (the child was stopped at 10); (c) count up from 5 to 9 and then up from 4 to 8; (d) count up n steps from a (2 steps from 3, from 4, and from 5; 3 steps from 4 and from 5, 4 steps from 5)³; and (e) count backward (from 7 and then from 15). A trial was judged to be incorrect if the counting sequence produced was incorrect; if the child used subvocal counting in trials (b), (c), (d), or (e); or if he or she did not stick to the rules given. The number of correct trials (maximum = 14) was used as the measure of the elaboration level.

Additions. A series of 10 additions was presented to the child: four ties (2 + 2, 3 + 3, 4 + 4, 5 + 5) and six additions corresponding to the items used in the “count up n steps from a ” task, but presented with the smaller addend first (2 + 3, 2 + 4, 2 + 5, 3 + 4, 3 + 5, 4 + 5) to allow a distinction between the counting-on and the counting-min strategies. For each item, the child was presented with a drawing of a collection of items representing the first operand, and tokens were at his or her disposal. The problem was presented orally (e.g., “Look, here are three cows; if four more come, how many cows will there be?”). The examiner recorded the child’s answer; the child’s use of the drawing, tokens, or fingers to count; and the counting strategy used (counting all, counting on, counting min, or noncounting strategies suggesting retrieval or guessing; Siegler, 1999).

General vocabulary. Finally, general vocabulary was assessed through the Vocabulary subtest of the EDEI-R battery (a test for 4- to 9-year-old children; Perron-Borelli, 1996). A series of 37 drawings was presented to the child, who was asked to name each of them.

Nonverbal intelligence. Nonverbal intelligence was assessed through the administration of Raven’s color progressive matrices (Raven, Court, & Raven, 1998).

Procedure

Each child was tested individually in a quiet room at school on four different occasions lasting 15–20 min each. In the first session, the following tasks were presented in this order: forward counting (first trial), N-fluency, elaboration of counting (count up to b , count up from a , count up from a to b), one-syllable word span, elaboration (counting backward, count up n steps from a), Corsi span, food and animal span, forward counting (second trial). In the second session, the child tackled the following tasks: category span, vocabulary subtest, nonword repetition, forward counting (third trial), reverse word span, and N-decision. Raven’s color progressive matrices were presented in the third session. The last session contained the additions, nonword repetition after 5 s and 10 s, and naming speed.

Results

Outline of the Results Section

The data from each individual task are analyzed first. Then, I come to the main point of this article that examines the links between the WM and the numerical measures. Finally, to illustrate the role of WM capacity on numerical development, the numerical

profiles of subgroups of children with above- and below-average WM capacity are compared.

The WM Tasks

As can be seen in Table 1, all the measures of WM capacity increased significantly between K2 (second year of kindergarten) and K3 (third year of kindergarten). To check the validity of the measures, several analyses were conducted. First, to test whether delay introduced in the repetition of nonwords had an impact on the child’s production, a mixed ANOVA was run with year as the between-subjects variable and delay (no delay, 5-s delay, or 10-s delay) as the within-subject variable. The main effects of year, $F(1, 78) = 17.26, p < .0001, \eta^2 = .18$, and delay, $F(2, 156) = 79.077, p < .0001, \eta^2 = .50$, were significant and interacted with one another, $F(2, 156) = 4.29, p = .015, \eta^2 = .05$. They showed that the difference between the two groups increased with the delay (difference of 0.33 in the immediate repetition and 1.05 in the 10-s delay), and that the sensitivity to the delay was larger in K2 (difference of 1.93 between the immediate and 10-s delay repetition) than in K3 (difference of 1.21; see Table 1 for the means and SD).

Second, to assess the impact of the processing component in the short-term memory tasks, I compared performance on the simple and complex tasks that used the same material (i.e., the word and reverse-word spans, and the food–animal and category spans). An ANOVA was run with year as the between-subjects factor and two repeated measures, the type of span (simple or complex) and the material (words or food–animals). The three main effects were all significant ($p < .0001$) and indicated that K3 children performed better than K2 children, $F(1, 78) = 20.82, \eta^2 = .18$; that complex spans were more difficult than simple spans, $F(1, 78) = 530.37, \eta^2 = .82$; and finally, that one-syllable word spans were more difficult than food–animal spans, $F(1, 78) = 86.97, p < .00001, \eta^2 = .53$. Two interactions were also significant. The Year $\times$ Complexity interaction, $F(1, 78) = 7.91, p = .006, \eta^2 = .09$, indicated that the decrease in performance between the simple and complex span tasks was greater in K2 than in K3 (difference of 1.99 and 1.56, respectively), which indicates that the capacity of the CE increased from K2 to K3 or that K3 children had greater processing efficiency to deal with the task. The interaction between complexity and material, $F(1, 78) = 131.15, p < .0001, \eta^2 = .63$, indicated that in the simple spans, the two materials did not differ, $t(79) = 1.00, ns$, means of 3.72 and 3.66 for words and food–animals, respectively, but did differ in the complex spans, $t(79) = 12.37, p < .0001$, means of 1.34 for the reverse span and 2.52 for the category span. In other words, repeating words in the reverse order was more difficult than repeating them by categories. Note that at this age, working memory capacity is so small that adding a processing component to a memory task often renders the task impossible. A considerable number of K2 children could not do the reverse span even with the two-word series, whereas nearly all the children could do the category span. This task is thus more appropriate for kindergarten children.

³ The task was explained in the following way:

This is a frog which jumps. If it jumps three times, it does *one, two, three*; if it jumps two times, starting at five, it does *five, six, seven*. Now it is your turn: the frog starts at three and jumps two times, so it does . . . ?

Table 1
Mean (SD) Performance for Each Task in K2 and K3

Task	K2 (n = 38)	K3 (n = 42)	t (df = 78)	p (one-tailed)	95% CI
Phonological loop					
Word span	3.56 (0.56)	3.87 (0.54)	2.46	.008	0.06, 0.55
Food–animal span	3.51 (0.56)	3.80 (0.53)	2.33	.011	0.04, 0.53
Nonword repetition	4.50 (0.97)	4.83 (0.89)	1.60	.056	–0.08, 0.75
Nonword repetition (5 s)	3.42 (0.94)	3.96 (0.93)	2.58	.006	0.12, 0.96
Nonword repetition (10 s)	2.57 (1.00)	3.62 (0.92)	4.88	.0001	0.62, 1.48
Naming speed	0.76 (0.18)	0.92 (0.25)	3.29	.001	0.06, 0.26
Visuo-spatial sketchpad					
Corsi	2.70 (0.78)	3.54 (0.76)	4.85	.0001	0.49, 1.18
Central executive					
Reverse-word span	0.92 (1.06)	1.71 (0.94)	3.52	.0001	0.34, 1.24
Category span	2.17 (0.93)	2.83 (0.66)	3.70	.0001	0.31, 1.02
Numerical skills					
Best counting	15.84 (9.55)	29.52 (15.51)	4.80	.0001	7.99, 19.37
SC counting	11.37 (7.48)	23.10 (11.96)	5.31	.0001	7.32, 16.13
N-fluency	21.24 (11.07)	40.57 (18.26)	5.79	.0001	12.66, 26.00
N-decision	14.0 (3.73)	16.51 (2.64)	3.42	.001	1.04, 3.96
Elaboration	1.89 (2.66)	5.09 (4.05)	4.22	.0001	1.69, 4.71
Additions	0.95 (1.58)	3.67 (2.88)	5.31	.0001	1.70, 3.74
Other measures					
Vocabulary	28.29 (3.72)	30.86 (3.99)	2.97	.002	0.85, 4.29
Raven matrices	11.13 (3.38)	15.55 (3.50)	5.80	.0001	2.90, 5.93

Note. CI = confidence interval of the difference.

Third, correlations (see Table 2) were calculated among the five simple verbal spans, the naming speed, the Corsi span, and the two complex span tasks. Performances on the five simple verbal span tasks correlated significantly with one another (except for the nonword 5-s repetition with both the nonword 0-s repetition and the food–animal span). These measures presented nonsignificant or only small correlations with the Corsi span that disappeared when age was partialled out. The naming speed had small but significant correlations, with the simple word tasks (except the nonword 0-s and 5-s repetitions), the Corsi span, and the two complex spans. Yet these correlations also turned out to be nonsignificant when controlling for age. Finally, the complex-span tasks were significantly correlated with one another as well as mildly correlated with the simple-span tasks for both the verbal and the visuo-spatial material.

To work with a smaller number of variables, three composite scores were computed on the basis of the correlation matrices and

theoretical considerations: (a) a PL composite score (the mean of the *z* scores for the word span, the animal–food span and the immediate nonword repetition span), (b) a rate of decay measure (the difference between the immediate and the 10-s delay nonword repetition), and (c) a CE measure (the mean of the *z* scores of the two complex spans). These measures, together with the Corsi span and the naming speed, were used to explain the performance on the numerical tasks. Table 3 shows the correlations between these measures.

The Numerical Tasks

All the measures showed a significant increase between K2 and K3 (see Table 1). As few data exist concerning the first numerical development in preschool, especially in new tasks such as the N-fluency or the N-decision, detailed descriptions of the performance are provided.

Table 2
Correlations Between the Cognitive Variables

Variable	1	2	3	4	5	6	7	8	9
1. Word span	—	.51***	.46***	.31*	.35***	.28*	.21	.45***	.44***
2. Food–animal span	.45***	—	.43***	.02	.25*	.22*	.25*	.40***	.50***
3. Nonword (0 s)	.42***	.38***	—	.21	.32*	.16	.17	.40***	.43***
4. Nonword (5 s)	.23*	–.10	.15	—	.48***	.02	.13	.34*	.36***
5. Nonword (10 s)	.26*	.13	.26*	.41***	—	.28*	.20	.44***	.57***
6. Naming speed	.17	.08	.06	–.12	.14	—	.33*	.38***	.30*
7. Corsi	.03	.07	.04	–.04	–.01	.13	—	.36***	.47***
8. Reverse word span	.34*	.28*	.33*	.22*	.31*	.21	.11	—	.63***
9. Category span	.35***	.41***	.37***	.27*	.48***	.13	.30*	.52***	—

Note. Simple correlations are presented above the diagonal; partial correlations (when controlling for age in months) are presented below the diagonal.
* $p < .05$. *** $p < .0001$.

Table 3
Correlations Between the Cognitive Indexes

Index	Matrices	PL	Rate decay	Naming speed	VSSP	CE
Age	.50***	.37***	-.23*	.43***	.54***	.53***
Matrices	—	.40***	-.23*	.33*	.40***	.46***
PL	.20	—	.01	.27*	.25*	.59***
Rate decay	-.16	.02	—	-.14	-.10	-.31***
Naming speed	.15	.13	-.05	—	.32*	-.38***
VSSP	.18	.04	-.01	.13	—	.45***
CE	.26*	.49***	-.30***	.20	.22	—

Note. Simple correlations are above the diagonal; partial correlations (when controlling for age) are below the diagonal. PL = phonological loop; VSSP = visuo-spatial sketchpad; CE = central executive.

* $p < .05$. *** $p < .0001$.

Counting. The best counting of K2 children was 15.84 on the average. In K3, it was 29.52. Similarly, the stable and conventional portion of the counting (SC-counting) sequence also increased from K2 to K3, going from 11.37 to 23.10.

N-fluency. N-Fluency increased with age. The K2 and K3 groups were similar in their ability to produce unit names (8.71 and 8.98), $t(39.5) = 2.01$, $p = .052$, $CI = 0.00, 0.53$; differed slightly in their ability to produce teens (3.16 and 5.50), $t(78) = 3.41$, $p < .001$, $CI = 0.97, 3.71$, and tens (3.26 and 4.38), $t(78) = 2.52$, $p = .014$, $CI = 0.24, 2.00$; and were very different in their ability to combine tens and units (6.05 and 21.36), $t(63.3) = 5.50$, $p < .0001$, $CI = 9.61, 21.00$. (Note that when the Levene test indicates a significant difference in the variance between the two groups, the degrees of freedom of the t test are not integers.) This reveals a development in the lexicon and, more importantly, in the mastery of combination in the verbal number system between K2 and K3.

N-decision. The N-decision task gave scores above chance, $M = 74\%$, $t(37) = 7.41$, $p < .0001$, in K2; and $M = 87\%$, $t(41) = 17.16$, $p < .0001$, in K3. An ANOVA calculated on the type of item (number or nonnumber words) and year indicated that the percentage of accuracy was significantly higher in K3 than in K2, $F(1, 78) = 12.88$, $p < .001$, $\eta^2 = .14$, and that it was easier for children to accept number words than to reject nonnumber words (means of 86.8% and 75.1%, respectively), $F(1, 78) = 11.51$, $p < .001$, $\eta^2 = .13$. The interaction between year and item, $F(1, 78) < 1$, was not significant.

Elaboration. Overall, K2 children performed worse than K3 children on elaboration tasks. This difference was observed on all the tasks used to assess the elaboration level. Indeed, only 17 K2 children, compared with 33 K3 children, twice counted correctly up to a , whereas 17 K2s and 5 K3s were unable to do this even once, $\chi^2(2, N = 80) = 11.49$, $p = .003$. When asked to count up from a number other than one, 7 K2s and 21 K3s did it correctly on two occasions, whereas 30 K2s and 17 K3s did not succeed at all, $\chi^2(2, N = 80) = 12.23$, $p = .002$. When required to count up from a to b , 5 K2 and 18 K3 children succeeded on both trials, whereas 31 K2s and 21 K3s never managed to do it, $\chi^2(2, N = 80) = 9.29$, $p = .01$. In the backward counting, only 4 K3 children completed the task successfully on two trials, whereas 9 K3s and only 1 K2 did it once, $\chi^2(2, N = 80) = 11.20$, $p = .004$. Finally, in counting up n steps from a , all but one child in K2 totally failed

at the task, whereas 11 K3 children succeeded in at least one of the trials (Mann Whitney, $U = 615$, $Z = -2.84$, $p < .005$).

Additions. On average, K2 children were able to solve correctly 9.5% of the additions, whereas K3 children could solve 36.7% of them. The problems solved correctly were usually tackled either by using mental strategies such as retrieval or guessing (56% of the trials), or by the counting-all strategy (44% in the whole sample, 42% in K2, 44% in K3). No other counting strategy was observed.

For the correct trials solved with a counting strategy (i.e., 44%), most of the time the children counted the pictures in the image and/or the tokens at their disposal (32% of the correct trials). Less often, they counted on their fingers (10%) or counted out loud without using any visible external support (4%). There were some differences between the two classes in the use of fingers (28% of correct answers in K2 and 10% in K3) and materials (14% in K2 and 32% in K3).

Klein and Bisanz (2000) observed that children's accuracy in nonverbal problems was strongly related to the representational set size, which, in the case of additions, corresponds to the correct sum. To see whether the same findings were replicated, a stepwise regression analysis was computed with percentage of accuracy of each addition as the dependent variable and sum of the problem, size of the first operand, size of the second operand, and tie status as potential predictors (all factors known to affect calculation performance in adults and school-age children; see Zbrodoff & Logan, 2005). Contrary to Klein and Bisanz, size of the second operand was the stronger predictor, explaining 55% of the variance, $F(1, 8) = 23.92$, $p < .0001$. Tie status was the second variable selected and improved the R^2 by .15, $F(1, 7) = 10.42$, $p = .014$. In other words, additions were completed more accurately if the second operand was small and if the two operands were equal. These two variables accounted for 70% of the variance in accuracy. The same results were obtained when only K3 performance was taken into account. But this time, size of the second operand explained 79% of the variance, $F(1, 8) = 30.23$, $p < .0001$; tie status added another 13% to the explained variance, $F(1, 7) = 11.42$, $p = .012$, for a total of 92% explained variance.

I then examined the parameters of the problem that were determinant for the selection of a resolution strategy. To that aim, the same analysis as described above was run on the percentage of correct additions solved by counting procedures (on K3 children only). The size of the second operand was the only significant predictor, $F(1, 8) = 31.13$, $p < .0001$. It accounted for 80% of the variance with a positive beta value (.89), which means that among accurate additions, the percentage of resolution through counting was increasing with the size of the second addend.

Correlations between numerical tasks. As can be seen in Table 4, the numerical measures were all significantly intercorrelated. As the best counting and SC-counting correlated strongly with one another and were derived from the same task, only the latter is considered below. The first factor of a principal component analysis run on the five numerical measures explained 68% of the overall variance with a loading of .89 for SC-counting, .90 for N-fluency, .67 for N-decision, .86 for elaboration and .79 for addition. This numerical factor (calculated on the basis of the loadings of each measure on this first factor) is used as a global

Table 4
Correlations Between the Numerical Variables

Variable	1	2	3	4	5	6
1. Best counting	—	.92**	.88***	.43**	.60**	.48**
2. SC-counting	.89**	—	.87**	.46**	.72**	.58**
3. N-fluency	.83**	.81**	—	.45**	.71**	.58**
4. N-decision	.28*	.31*	.34*	—	.47**	.47**
5. Elaboration	.47**	.62**	.60**	.34*	—	.63**
6. Additions	.30*	.41**	.40**	.33*	.51**	—

Note. Simple correlations are presented above the diagonal; partial correlations (when controlling for age in months) are presented below the diagonal.

* $p < .05$. ** $p < .001$.

measure of numerical development in the next section when considering the link between numerical and WM capacities.

WM Capacity and Performance on the Numerical Tasks

In order to detect which WM component plays a significant role in the first numerical development, partial correlations were calculated between the children's performance on each numerical task and their WM capacities. These correlations were measured after controlling for the child's age (expressed in months) and nonverbal intelligence (raw score on Raven's matrices). The same analysis was also performed on the Vocabulary test.

As Table 5 shows, the PL capacity was significantly correlated with SC-counting, N-fluency, elaboration and the numerical factor. Elaboration only was moderately correlated with naming speed. VSSP capacity did not correlate significantly with the performance on any of the numerical tasks. More strikingly, all numerical measures were significantly correlated with the CE capacity. Furthermore, this was not due to the fact that several younger children were not able to deal with one of the CE tasks (i.e., the reverse span). Indeed, if only the performance on the category span is considered, which was very appropriate even for the younger children, the correlations with the CE are equivalent if not stronger.

In line with Fuson (1982)'s idea that fingers and objects are used to keep track of the counting, correlations were also calculated between WM measures and the number of additions correctly solved without any external support, in K3 children only (as the accuracy rate was small in K2). Simple correlations indicated that accuracy with no external support was higher in children who had good capacities in VSSP, $r_{(40)} = .32$, $p = .04$, and CE ($r = .67$, $p = .02$), whereas the correlations with PL ($r = .15$), delay ($r = .05$) and naming speed (.21) were not significant. However, when age and nonverbal intelligence were partialled out, the correlations with VSSP and CE were no more significant, $r_{(38)} = .29$ and .28, respectively.

Finally, vocabulary was significantly correlated with measures of the PL, rate of decay, and naming speed, but also, and even more strongly, with the CE measures.

Next, using multiple regressions, I determined which of these WM measures was the strongest predictor of performance in the numerical and vocabulary tasks. I used a model that first forced the entry of age (in months) and raw score on matrices (Step 1) and then used a stepwise selection method to examine which of the five

other cognitive measures (PL, decay, naming speed, VSSP, CE) was the best predictor of the dependent variable. As can be seen in Table 6 (Step 2), the measure of the CE was the best predictor of performance for each of the five numerical tasks, as well as the global numerical factor, for which it increased the explained variance by 18%. The next significant predictor selected (Step 3) was the measure of PL for SC-counting and for elaboration. A small influence of naming speed was also found for elaboration. Finally, performance on vocabulary was also best predicted by the CE measure (increasing the explained variance by 21%), with a smaller contribution of naming speed (accounting for an additional 5% of explained variance).

Although PL capacities significantly correlated with all these dependent variables (except the N-decision and the addition), it is possible that it was not retained as a significant predictor in the analyses above because measures of the CE used verbal materials. Accordingly, these measures involved the need to store (and manipulate) verbal material and thus, in a sense, are already measuring the PL capacities. Given these considerations, another series of multiple regression analyses was run, this time forcing the entry of age and nonverbal intelligence, and then of the PL measures (Step 2' in Table 6). The other four cognitive predictors were then subjected to a stepwise selection procedure. These supplementary analyses showed that the PL made a significant contribution to SC-counting, N-fluency, and elaboration, and globally contributed to the numerical factor but not to N-decision and addition. Importantly, forcing the entry of PL did not prevent the CE measure from being selected by the stepwise procedure (see Step 3') for all the numerical tasks. For the numerical factor, the PL explained 10% of the variance and the CE still added another 10% of explained variance. The same observation was also true for vocabulary. In this case, PL explained 7% of the variance and CE added another 14% of explained variance.

To examine the effects of CE capacities on the numerical and vocabulary development of children more concretely, children were divided into four age groups (47–53, 54–59, 60–65, and > 66 months old). I considered them to have low or high CE capacity if their CE score was lower or higher, respectively, than the median of their group age. ANOVAs computed with this CE group factor and year indicated that the two CE groups did not differ in terms of age, $F(1, 76) < 1$, and only mildly on nonverbal intelligence:

Table 5
Correlations Between the WM Capacities and the Performance in the Numerical and Vocabulary Tests (After Controlling Out for Age and Nonverbal Intelligence)

Variable	PL	Rate of decay	Naming speed	VSSP	CE	Category span
SC-counting	.47***	-.17	.14	.12	.53***	.58***
N-Fluency	.38***	-.14	.20	.14	.51***	.55***
N-Decision	.22	-.12	.08	-.05	.40***	.39***
Elaboration	.48***	-.08	.31*	.23	.57***	.61***
Additions	.12	.02	.09	.13	.30*	.37***
Numerical factor	.45***	-.13	.22	.16	.61***	.61***
Vocabulary	.34*	-.24*	.37*	.20	.56***	.49***

Note. $df = 75$. PL = phonological loop; VSSP = visuo-spatial sketchpad; CE = central executive.

* $p < .05$. *** $p < .0001$.

Table 6
Results of the Regression Analyses

Step	Adjusted R^2	ΔR^2	F	$df1$	$df2$	p
DV = SC counting						
1. Age and matrices (forced)	.35	.37	22.71	2	77	.0001
2. CE	.50	.15	23.57	1	76	.0001
3. PL	.53	.03	5.97	1	75	.017
2'. PL (forced)	.48	.12	18.74	1	76	.0001
3'. CE	.53	.06	10.10	1	75	.002
DV = N-fluency						
1. Age and matrices (forced)	.38	.40	25.44	2	77	.0001
2. CE	.51	.13	21.35	1	76	.0001
2'. PL (forced)	.45	.07	10.18	1	76	.002
3'. CE	.52	.07	11.64	1	75	.001
DV = N-decision						
1. Age and matrices (forced)	.16	.19	8.74	2	77	.0001
2. CE	.28	.12	13.01	1	76	.001
2'. PL (forced)	.19	.03	3.17	1	76	<i>ns</i>
3'. CE	.27	.09	9.34	1	75	.003
DV = Elaboration						
1. Age and matrices (forced)	.28	.30	16.03	2	77	.0001
2. CE	.48	.20	30.56	1	76	.0001
3. PL	.51	.04	5.84	1	75	.018
4. Naming speed	.53	.02	4.05	1	74	.048
2'. PL (forced)	.44	.15	20.39	1	76	.0001
3'. CE	.53	.09	14.38	1	75	.0001
DV = Additions						
1. Age and matrices (forced)	.31	.32	18.47	2	77	.0001
2. CE	.34	.04	5.48	1	76	.02
2'. PL (forced)	.30	.00	<1	1	76	<i>ns</i>
3'. CE	.34	.04	4.88	1	75	.03
DV = Numerical Factor						
1. Age and matrices (forced)	.44	.46	32.11	2	77	.0001
2. CE	.62	.18	38.09	1	76	.0001
2'. PL (forced)	.53	.10	16.57	1	76	.0001
3'. CE	.63	.10	21.18	1	75	.0001
DV = Vocabulary						
1. Age and matrices (forced)	.21	.23	11.39	2	77	.0001
2. CE	.41	.21	27.75	1	76	.0001
3. Naming speed	.46	.05	7.85	1	75	.006
2'. PL (forced)	.27	.07	7.38	1	76	.0001
3'. CE	.41	.14	18.65	1	75	.0001

Note. Age (in months) and performance in the matrices are forced to entry in Step 1. Phonological loop (PL) is forced to entry in Step 2' (after age and matrices). Dependant variables (DV) are the performance in the numerical tasks and in the vocabulary subtest. Adjusted R^2 , ΔR^2 , and F statistics for ΔR^2 are reported. CE = central executive.

matrices, $F(1, 76) = 4.22$, $p = .043$, $\eta^2 = .05$ (see Table 7). Of course, year had a significant effect on age, $F(1, 76) = 179.79$, $p < .0001$, $\eta^2 = .70$, and on the raw scores on matrices, $F(1, 76) = 37.50$, $p < .0001$, $\eta^2 = .31$. More important, year and CE group influenced significantly (all $ps < .001$) the performance on all the numerical measures as well as on vocabulary: SC-counting, $F(1, 76) = 37.50$, $\eta^2 = .33$, and $F(1, 76) = 24.76$, $\eta^2 = .25$, for the main effects of year and CE group, respectively; N-fluency, $F(1, 76) = 44.37$, $\eta^2 = .37$, and

$F(1, 76) = 24.18$, $\eta^2 = .24$; N-decision, $F(1, 76) = 14.64$, $\eta^2 = .16$, and $F(1, 76) = 13.71$, $\eta^2 = .15$; elaboration, $F(1, 76) = 25.50$, $\eta^2 = .25$, and $F(1, 76) = 25.50$, $\eta^2 = .25$; additions, $F(1, 76) = 31.57$, $\eta^2 = .29$, and $F(1, 76) = 9.49$, $\eta^2 = .11$; vocabulary, $F(1, 76) = 10.93$, $\eta^2 = .13$, and $F(1, 76) = 17.70$, $\eta^2 = .19$. These two factors had additive effects. The contrast between these four groups is illustrated in Figure 1 with the corresponding z scores. This figure shows the comparable effects of year and CE group on the performance; the profile of

Table 7
Mean (SD) Performance for Each Task in the Low and High CE Groups on K2 and K3

Measures	Low CE, K2 (n = 20)	High CE, K2 (n = 18)	Low CE, K3 (n = 24)	High CE, K3 (n = 18)
Age	52.55 (3.91)	53.56 (3.87)	65.29 (4.11)	64.39 (3.63)
Raven matrices	10.0 (3.55)	12.39 (2.50)	15.25 (3.50)	15.94 (3.55)
Numerical skills				
SC-counting	6.90 (5.21)	16.33 (6.46)	18.62 (10.12)	29.06 (11.86)
N-fluency	14.95 (7.86)	28.22 (9.98)	33.50 (17.06)	50.00 (15.67)
N-decision	12.65 (4.04)	15.50 (2.77)	15.58 (2.90)	17.72 (1.64)
Elaboration	0.60 (1.13)	3.33 (3.09)	3.33 (2.90)	7.44 (4.23)
Additions	0.40 (0.82)	1.56 (1.98)	2.88 (2.33)	4.78 (3.21)
Vocabulary	26.15 (3.47)	30.67 (2.30)	29.95 (4.26)	32.06 (3.35)

Note. CE = central executive.

K3 children with low CE capacity was very similar to the one obtained by K2 children with high CE capacity.

Discussion and Conclusion

Working memory plays an important role in many cognitive processes. In the numerical domain, both the CE capacity and, to a lesser extent, the PL capacity have already been shown to contribute to arithmetical skills in both adults and primary-school children. The VSSP has also occasionally been shown to make a contribution (e.g., McKenzie et al., 2003). Not much is known about the role of WM in the very first steps of learning arithmetic. Klein and Bisanz (2000) argued that preschoolers' performance in arithmetic is mainly restricted by their limited WM capacity, but they did not test this capacity directly. Rasmussen and Bisanz (2005) assessed the PL, VSSP, and CE capacities of preschoolers and pointed out that the various components of working memory are differentially involved in addition, depending on the presentation format of the problem: The CE capacity was significantly correlated with accuracy in verbal problems, whereas the VSSP capacity was correlated with accuracy in nonverbal problems.

Even less is known about the memory constraints of the counting development, although counting skills are the bases for the development of calculation strategies and are a good predictor of mathematical school achievement years later (Krajewski & Schneider, 2008).

In this study, the role of WM in early arithmetical development was examined by submitting 4- and 5-year-old preschoolers to tests assessing WM (i.e., PL, VSSP, and CE), as well as to simple additions. Unlike Rasmussen and Bisanz (2005), who imposed a particular method of solving problems (by presenting objects and asking for an answer involving objects or by presenting verbal problems and requiring a verbal answer), I opted for a more open situation in which the problems were presented both visually and orally and tokens were available to help the children perform the addition. With this approach, the children were left as much freedom as possible to use the solving method that suited them best. I also tested the children's counting skills and number vocabulary, as these constitute one of the bases for developing the strategies necessary to perform additions. The contribution of WM capacities to this development has never been explored before.

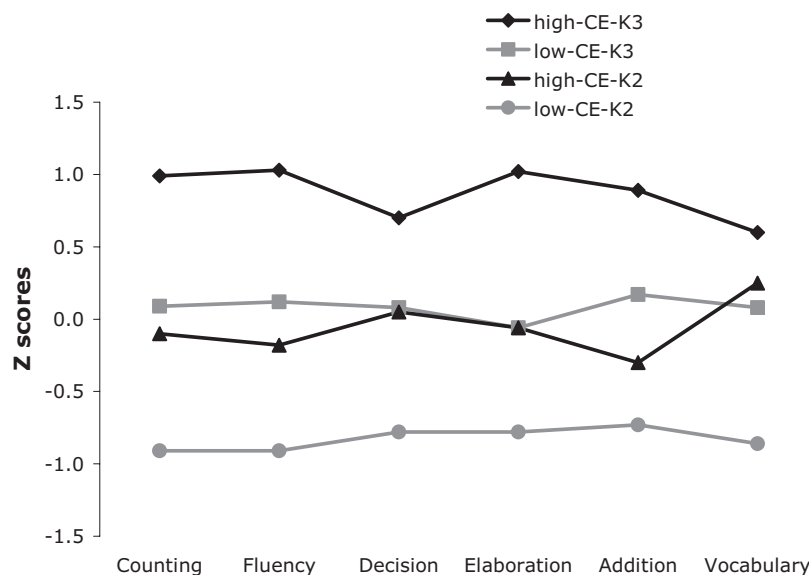


Figure 1. Z scores for the high and low central executive (CE) groups in K2 and K3 for all the tasks.

The WM Tasks

The results of the WM tasks indicate that performance increased with age; children were sensitive to the retention delay when asked to repeat nonwords and to the processing load required in complex span tasks, and K2 children were more sensitive to these two factors than K3 children. The various measures of the PL were significantly correlated with one another, and good correlations were also obtained between the two CE measures. CE measures for such a young population are very scarce. The reverse word span appeared to be too difficult for many 4-year-olds, but the category span was perfectly appropriate and certainly deserves more attention in future work with very young children.

The pattern of intercorrelations among these WM components showed some significant correlations between the measures of the PL and of the VVSP that disappeared when age was controlled. However, measures of the CE were significantly correlated with those of the PL (even after controlling for age) and more moderately with those of the VSSP (although one of the two correlations did not reach the significance threshold when age was partialled out).

In the study of Rasmussen and Bisanz (2005), simple intercorrelations between the WM measures were all significant in preschoolers. In 6- and 7-year-old children, Gathercole and Pickering (2000) used a full WM battery with four tasks to measure the VSSP, six for the PL, and three for the CE. Out of the 24 simple correlations calculated between the PL and the VSSP measures, only 3 were significant; 5 out of 18 significant correlations were measured between the PL and the CE measures, and 9 out of 12 between the VSSP and the CE measures. In young adults, measures of the VSSP (similar to the Corsi) mildly correlated with measures of the PL (digit and word spans, $r = .26$ and $.30$, $n = 53$), but measures of both slave systems were significantly correlated with those of the CE (reading and computation spans, $r = .53$ and $.43$ for the VSSP, r range from $.35$ to $.50$ for the PL; Oberauer, Süß, Wilhelm, & Wittman, 2003).

In other words, in preschoolers as well as in older age groups, intercorrelations between the different WM components are significant. The correlation between the two passive stores (VSSP and PL) is usually weak, and in this study it even disappears when age is partialled out, whereas measures of both slave systems significantly correlate with measures of the CE.

The Numerical Tasks

An increase in performance from K2 to K3 was also observed on the numerical tasks. On average, children in K2 counted up to 11 correctly, whereas children in K3 were able to count up to 23. This is in accord with Fuson et al.'s (1982) data. When the order in which numbers were named was ignored, K3 children had better number fluency than K2 children, especially in producing tens-unit structures. This suggests a better mastery of the combinatory rules of the numerical system. In a number decision task, K3 children again performed better than K2 children. In tasks testing the level of elaboration of the counting string, the differences were striking: Nearly 90% of K3 children could count up to a at least once, more than half of them could count up from a and count up from a to b , and more than a quarter of them could also count backward and count up n steps from a . By contrast, less than half of the K2

children could count up to a at least once, and more than 80% of them completely failed all the other items.

A large difference also appeared in the addition task, with 10% accuracy in K2 and 37% in K3. To perform additions correctly, the children used mental strategies 56% of the time and a counting-all strategy in the other cases. This replicates the findings of Siegler and Jenkins (1989), who reported that the most frequent strategies used for addition by 4- and 5-year-old children were counting-all and retrieval.⁴ Only the counting-all algorithm was used in the problems solved correctly by counting. Children usually used the images or tokens at their disposal, rather than their fingers, which suggests that fingers are mainly used when nothing else is available to represent the operands.

Importantly, these numerical measures were highly intercorrelated. A similarly strong relationship between counting and arithmetical skills has previously been reported by Donlan, Cowan, Newton, and Lloyd (2007) in a study with language-impaired children; weak counting abilities have been reported in children with low mathematical skills (Hitch & McAuley, 1991), and counting skills in preschool are a good predictor of mathematical school achievement years later (Krajewski & Schneider, 2008). This study is the first to report a strong link in typically developing preschool children.

Relationships Between WM and Numerical Tasks

The main purpose of this study was to evaluate the contribution of WM to numerical development and to specify the subcomponent(s) of the WM that made the greatest contribution to counting and arithmetical skills. As far as counting is concerned, PL was expected to be important, because many studies have pointed to the role of the PL in vocabulary acquisition (e.g., Gathercole & Baddeley, 1993; Masoura et al., 2004) and disruption of the PL has been shown to generate counting errors in adults (Healy & Nairne, 1985; Logie & Baddeley, 1987). The CE was also a good candidate: It has been found to be less performant in children with specific language impairments (Gathercole & Pickering, 2000) and to predict vocabulary skills (Gathercole et al., 2005). Concerning the level of elaboration of the counting sequence, I predicted that children with good CE capacity would have a better development as most of the tasks require monitoring, a function dedicated to the CE.

Partial correlations controlling for the child's age and nonverbal intelligence showed that, as predicted, counting skills were significantly correlated with both the PL and the CE measures, but not with the VSSP measure. More precisely, the PL capacity correlated significantly with SC-counting, N-fluency, and elaboration, whereas CE capacity was significantly correlated with these three measures as well as with N-decision. Regression analyses showed that CE was the best predictor of all the tasks involving numerical vocabulary (N-fluency, N-decision) and counting (SC-counting, elaboration). This was not simply a byproduct of the fact that the measures of the CE required the storage of verbal information:

⁴ In Siegler and Jenkins's (1989) study, guessing occurred in only 2% of the cases. On the basis of these data, one can probably infer that in my study, most of the problems that were solved mentally and led to a correct response used retrieval rather than guessing.

Even after controlling for PL capacity, as well as age and nonverbal intelligence, CE still explained a significant proportion of the variance (between 6% and 10%).

This observation held not only for numerical vocabulary, but also for general vocabulary. Although the PL capacity, the rate of decay of verbal information, and the naming speed were all significantly correlated with general vocabulary, CE was the best predictor. It accounted for 14% of the variance, after controlling for age, nonverbal intelligence, and PL capacity. An impressive number of studies have shown that measures of the PL correlate with vocabulary skills in children (e.g., Avons, Wragg, Cupples, & Lovegrove, 1998; Gathercole, Service, Hitch, Adams, & Martin, 1999; Majerus, Poncelet, Greffe, & Van der Linden, 2006). These authors have hypothesized that PL capacity is a causal factor in the development of vocabulary, and some computational models of word learning based on PL capacity have been put forward (e.g., Jones, Gobet, & Pine, 2007). However, the present study found that the link with CE was even stronger than the link with PL, although this component of the WM system has been largely neglected in studies of vocabulary development. These results are in line with Gathercole et al.'s (2005) findings that the vocabulary of 8-year-olds was more closely related to CE capacity than to PL capacity. Gathercole et al. suggested that the association between PL and vocabulary might be strong at 4 years of age and then decline to a nonsignificant level at age 8. Yet in this study I have shown that even at the age of 4 or 5, measures of CE are better predictors of vocabulary than measures of PL. The CE should be thought of as a system that is able to encode and retrieve information not only from the slave systems but also from long-term memory (Baddeley, 1996). Its role in vocabulary acquisition, including the number words sequence, should certainly be further explored in the future.

A variety of hypotheses were explored for addition. The literature on primary-school children strongly supports a link between CE capacity and performance in addition, whereas in preschoolers, the way in which the problem is presented (and the response given) has been shown to be of particular importance. Rasmussen and Bisanz (2005) have shown a specific role for the VSSP when nonverbal problems are involved, but a main role for the CE when verbal problems are used. In the present study, problems were presented both verbally and with pictures of objects, and tokens were available to the child to elaborate his or her answer. This was done to allow the child as much freedom as possible to choose the way to solve the problem. The results showed that performance in additions was significantly correlated with CE capacity, but not with PL or VSSP capacity. This is in line with Rasmussen and Bisanz's (2005) results on preschoolers answering verbal problems but conflicts with their results for nonverbal problems. This suggests that when they could choose their own method of tackling the problem, children did not generally use mental models based on object files. Of course, it is possible that the lack of a significant correlation between VSSP and accuracy was due to the fact that only one measure of VSSP capacity was employed, whereas there were two for CE capacity. However, Rasmussen and Bisanz (2005) also used only one measure of VSSP capacity (the same one as used here). Further study of the possible contribution of the VSSP to solving simple arithmetic problems in preschoolers might prove fruitful.

There were some differences between the present results and those reported previously by Rasmussen and Bisanz (2005) and Klein and Bisanz (2000). First, the accuracy rate observed here for 5-year-old children was higher than that reported by [Rasmussen and Bisanz \(2005\)](#) for either verbal or nonverbal problems (28% and 23%, respectively). This is probably due to differences in the presentation of the problem and associated WM constraints. In the present study, the first operand was visible (i.e., objects corresponding to the first operand had been drawn in a picture and were in full view of the child), whereas in the study by Rasmussen and Bisanz, it had to be held in WM (either in the PL for verbal problems or in the VSSP for nonverbal problems). This may have led to lower accuracy.

In Klein and Bisanz's (2000) study, accuracy on the different problems was strongly related to the representational-set size of the problem, which, in the case of additions, corresponded to the sum. However, in the current study, the size of the second operand was the best predictor of children's performance. Again, this discrepancy can be accounted for by differences in the presentation format of the problem and associated WM constraints. Klein and Bisanz placed an array of tokens on a mat and covered them; some tokens were then added to or subtracted from the set. So the child had to retain in WM the representation of the first addend (covered and thus invisible) and the modification of that array by the addition or subtraction of the second one. For addition, the child thus had to hold a representation of a set corresponding to the sum of the two arrays (i.e., the representational-set size) in his or her WM. However, in my study the first addend was always visible to the child and so did not need to be held in WM. Only the second addend (presented orally) had to be kept in WM, and it was the size of this second operand that was the best predictor of success in the addition. So, again, my results suggest that children's performance on addition tasks is constrained by the WM load in the specific situation used.

Finally, the WM constraints of the problem also had an impact on the solving strategies used by the children. Considering only the cases where the correct answer was achieved, the frequency of counting strategies (as opposed to mental ones) increased as the size of the second operand increased. In other words, children tended to favor the counting strategy and, more precisely, the counting-all strategy when the size of the operand that needed to be kept in WM was large. External supports, such as tokens, pictures, or more rarely fingers, were often used to help the child keep track of the counting when the demands on WM were great (see Fuson, 1982). The use of these external supports was more frequent in children with low VSSP and CE capacities. No significant correlations with PL measures were obtained, contrary to the results of [Noël et al. \(2004\)](#). Yet, in that study, older children (i.e., Grade 1 children) were tested and problems were presented in the Arabic code.

As correlations among numerical tasks were all significant, a global numerical factor was considered. Scores in this factor were significantly associated with measures of the PL capacities but even more importantly with the CE capacities.

Let us also note that naming speed was not a strong predictor of the numerical development (except for some contribution to the level of elaboration of the counting). This is in agreement with a recent study by Willbruger, Fussenegger, Moll, Wood, and Land-

erl (2008), which showed that dyscalculia was not associated with deficit in rapid naming.

In summary, children's performance on general and numerical vocabulary tasks, as well as on addition, were strongly predicted by their CE capacity (and this was not due to the fact that several younger children were not able to deal with the reverse span, as just the performance on the category span led to equivalent if not stronger correlations). Measures of the PL were also significantly linked to the child's numerical and general vocabulary, but their predictive power was weaker than that of the CE measures. In fact, measures of the PL capacity only made a specific contribution to explaining the length and level of elaboration of the counting string. VSSP capacity was not a significant predictor of any of the test scores. Using a simple median split to compare children with low and high CE capacity gave highly significant differences on all the numerical tasks and the vocabulary task. The effect of CE group was as important as the effect of class year and was additional to that effect. Therefore, the simple use of CE tests can allow the slow learners in the class, who might need special attention, to be identified. Prevention programs have been tested on children who are at risk of school failure due to their low socioeconomic status (Griffin, 2007) and have shown positive results. The results reported here identify another risk factor that should be taken into account. Further work should examine whether numerical stimulation of children with low CE capacity might be effective in compensating for their developmental lag and allowing them to enter primary school with a numerical background comparable to that of their classmates and sufficient for them to follow normal mathematics classes.

Alternatively, training memory in children with poor CE capacities (as proposed by Klingberg et al., 2005) might be even more relevant as this would probably lead to beneficial effects not only in learning mathematics but also in other learning domains.

My results thus confirm the important role of WM constraints, especially as regards the CE component, on the child's counting and arithmetical development.

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